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Key Points:
- When radiative interactions are suppressed, the global-mean eddy kinetic energy increases by ~6%
- Radiative cooling coincident with positive temperature anomalies destroys eddy available potential energy and eddy kinetic energy
- The energetic perspective provides a useful way to understand the effect of radiative heating on the general circulation

Supporting Information:
Supporting Information may be found in the online version of this article.

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Why Does Atmospheric Radiative Heating Weaken Midlatitude Cyclones?

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Abstract Recent work has indicated that atmospheric radiative heating reduces the kinetic energy of large-scale eddies in the midlatitudes. However, a physical mechanism that connects radiation to the midlatitude eddy kinetic energy is still uncertain. Using a high-resolution general circulation model we perform an experiment in which the radiative cooling profile at each model time step is overwritten with the climatological mean, computed from a control simulation. This approach separates the mean and transient effects of radiative heating on the extratropical circulation. We find that, when radiative heating is fixed, the globally-averaged eddy kinetic energy is enhanced by ~6%. We show that thermal radiation dampens temperature anomalies near the surface and tropopause in low-pressure systems, destroying eddy available potential energy and eddy kinetic energy. We identify this as a possible mechanism by which atmospheric radiative heating weakens midlatitude cyclones.

Plain Language Summary The atmosphere can be differentially heated through the absorption of visible and infrared radiation by water vapor and clouds. This process of radiative heating is thought to promote the formation of tropical cyclones, but its impact on midlatitude cyclones—storms characterized by the passage of warm and cold fronts—is less clear. Here we show that, unlike tropical cyclones, radiative heating weakens midlatitude storms. We argue that radiative cooling in the warm sector, and radiative heating in the cold sector of midlatitude cyclones, reduces the temperature gradient which is critical for the development of these weather systems. This study underscores an essential difference between tropical and midlatitude storms: in the tropics, temperature gradients are small, but at higher latitudes, temperature gradients are the drivers of storm formation.

1. Introduction

A recent study has demonstrated that radiative heating of the atmosphere reduces the kinetic energy of large-scale eddies (Schäfer & Voigt, 2018). This is a counterintuitive result because, in the first place, radiation is destabilizing in one-dimensional vertical convection: a cloud absorbing longwave radiation will be heated at its base and cooled at its top, furthering the process of convection. In the second place, there is a large body of literature showing that radiative feedbacks promote tropical cyclone formation, particularly at its earliest stages (Carstens & Wing, 2020; Hsieh et al., 2023; Muller & Romps, 2018; Ruppert et al., 2020; Smith et al., 2020; Wang et al., 2019; Wing, 2022; Wu et al., 2021; Zhang et al., 2021). Why midlatitude cyclones should have the opposite response to tropical cyclones is not obvious; and interpreting the results of Schäfer and Voigt (2018) is complicated by their method for isolating the effect of radiation (Keshtgar et al., 2023).

In addition to Schäfer and Voigt (2018), who analyzed both total radiative heating and cloud radiative heating, a number of studies have focused on the effect of cloud radiative heating, but have found conflicting results (Grise et al., 2019; Keshtgar et al., 2023; Li et al., 2015; Voigt et al., 2023). These studies employed a variety of methods, including turning cloud radiative heating on and off (Li et al., 2015), overwriting cloud properties with those from 1 year of a control simulation (Grise et al., 2019), and passing only cloud radiative heating into the dynamical core (Keshtgar et al., 2023; Voigt et al., 2023). One caveat of the method of Schäfer and Voigt (2018) and Li et al. (2015) is that by turning radiative heating on and off there are substantial changes to the background state in which the cyclones are developing (Keshtgar et al., 2023). We argue that it is preferable to separate the mean and transient effects of radiative heating on the general circulation, while keeping the total input of heat from radiation constant. We demonstrate a method for doing so here.

Because we are quantifying the vigor of large-scale eddies with their kinetic energy, it is convenient to view the general circulation in terms of its flows of energy. Lorenz (1955), who originated the idea of available potential

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energy, provides a widely-used framework for understanding these flows of energy, which we summarize as follows. The radiative imbalance from equator to pole generates mean available potential energy, which is transformed into eddy available potential energy and eddy kinetic energy through baroclinic instability; the eddy kinetic energy then maintains the zonal-mean kinetic energy against dissipation by friction through barotropic processes. If the atmosphere were adiabatic and frictionless, the sum of available potential and kinetic energy would be conserved: energy would only be transformed from one type to the other. Diabatic processes (radiation, condensation of water vapor, or conduction from the surface) affect only the generation of available potential energy; and the kinetic energy is altered by diabatic heating indirectly, through its effect on available potential energy.

The purpose of this study is to clarify the effect of variations of atmospheric radiative heating on the extratropical circulation, using the framework put forward by Lorenz (1955). We show, in an experiment where the radiative cooling profile is overwritten with the climatological mean from a control experiment, that synoptic-scale eddies are more vigorous when radiative feedbacks are suppressed. We find that radiative cooling near the surface and tropopause in low-pressure systems destroys eddy available potential energy by removing temperature anomalies. We propose this as a possible mechanism by which radiative heating weakens midlatitude cyclones.

2. Methods

The Geophysical Fluid Dynamics Laboratory High-Resolution Atmosphere Model (HiRAM; see Zhao et al., 2009) is used to determine the effect of atmospheric radiative heating on midlatitude cyclones. The model has ∼0.5° horizontal resolution and 32 vertical levels. The (relatively) high spatial resolution of HiRAM is advantageous because the storm tracks in climate models are better simulated at higher resolutions than coarser ones (Colle et al., 2013). At the lower boundary the model is forced by monthly-mean observed sea surface temperatures and sea ice from the HadISST data set (Rayner et al., 2003) between 1986 and 2005. Land surface temperatures are interactive and respond to surface radiative fluxes.

The experiments used in this study were first described by Zhang et al. (2021). In the control simulation the radiative transfer calculation is interactive, and the radiative heating and cooling rates at each time-step respond to the instantaneous conditions in the column at that time (e.g., specific humidity and cloud cover). In a perturbation experiment, referred to as "climatological radiation," the model is rerun, but at each time-step the total radiative heating or cooling rate (longwave plus shortwave) is overwritten with the monthly mean, calculated from the final 20 years of the control run, and interpolated to the current time-step. The virtue of this approach is that the time-mean radiative cooling rates are identical in the control and climatological radiation experiment. This stands in contrast to other experiments which make clouds transparent to radiation (set the cloud radiative heating rates to 0), and therefore introduce a disparity in the total heat input from radiation into the system. In the climatological radiation experiment, the radiative heating and cooling rates are prescribed from the control climatology at all but the top two model levels; radiation at upper levels is left interactive to ensure that surface fluxes exactly balance outgoing radiation, and there is no unusual surplus or deficit of energy in the atmosphere. A related experiment, in which the radiative cooling rates are prescribed only in the troposphere, is described in Section 3.3. We note that only the atmospheric radiative cooling rates are overwritten in these experiments and at the land surface the radiative fluxes are interactive.

For the analysis it is necessary to calculate mean and eddy quantities related to available potential and kinetic energy. As discussed by Oort (1964), eddies can be defined from the space, time, or mixed space-time domains. Here we consider transient eddies, that is, departures from the time mean. Therefore the eddy kinetic energy is given by

$$K_E = \frac{1}{2g} \int \overline{[u'^2 + v'^2]} dp,$$

where the overbar and bracket represent the time and zonal means, respectively, and the prime indicates the difference from the monthly mean (Shaw et al., 2022).

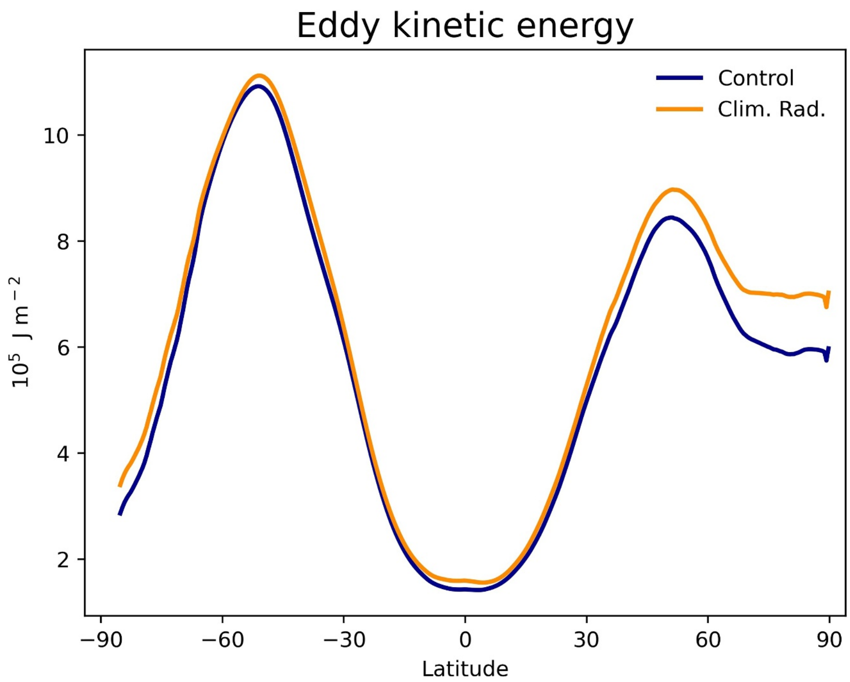

Figure 1. The vertically-integrated zonal-mean eddy kinetic energy for the control (blue), where the radiative heating rate responds to instantaneous conditions of moisture and cloud cover, and the climatological radiation experiment (orange), where the radiative heating rate is fixed to the monthly mean.

3. Results

3.1. Eddy Kinetic Energy

The vertically-integrated zonal-mean eddy kinetic energy of the control and climatological radiation experiment is reported in Figure 1. When radiative interactions are suppressed, the eddy kinetic energy is enhanced in the midlatitudes. This is consistent with the findings of Schäfer and Voigt (2018). In general, the zonal-mean vertical structure of eddy kinetic energy is characterized by two teardrop lobes of high eddy kinetic energy located at ∼40 N/S, with maxima near the tropopause. The change of eddy kinetic energy between the control and climatological radiation experiment is concentrated at upper levels in the troposphere (Figure S1 in Supporting Information S1), and in the midlatitudes can be thought of as intensifying the eddy kinetic energy where it is largest. We note there is some change of the eddy kinetic energy observable in the tropics, which may influence eddies in the midlatitudes. However, the mean tropical circulation and temperature profile are nearly constant between the control and climatological radiation experiment (Zhang et al., 2021).

3.2. Generation of Eddy Available Potential Energy

If the atmosphere were adiabatic and frictionless, the sum of available potential and kinetic energy would be conserved, and the two forms of energy could only be converted from one to the other. It is through diabatic processes (radiative heating or cooling, latent heat release, sensible heat transfer) that available potential energy is generated; and the kinetic energy can only be altered by diabatic heating through its indirect effect on available potential energy. The generation of available potential energy can be partitioned into mean and eddy components. Because the time-mean radiative cooling profile is constant between the control and climatological radiation experiment, it is only the generation of eddy available potential energy that is directly altered in the climatological radiation run. All accompanying changes to the flows of energy in the atmosphere must be responses to this initial perturbation. An expression for the globally-averaged generation of eddy available potential energy is given by Oort (1964):

$$G(P_E) = \int \gamma \overline{\left[T' Q'\right]} dm,$$

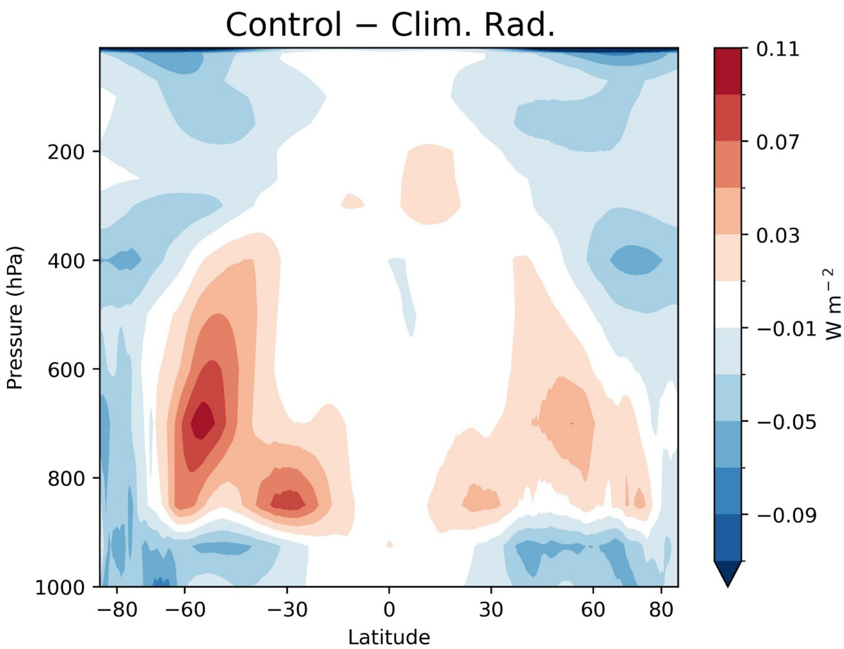

Figure 2. The difference of the generation of eddy available potential energy due to radiation between the control and climatological radiation experiment. In the midlatitudes, radiative interactions generate eddy available potential energy at mid-levels but destroy it near the surface and tropopause.

where $Q' = c_p \dot{T}'$ is the departure of the heating rate from the zonal mean, dm is an increment of mass, and $\gamma \propto \left(\frac{\theta}{T}\right)^2$ is a term arising from the derivation of available potential energy from the thermodynamic equation (see Oort, 1964). The interpretation of $G(P_E)$ is straightforward: when T' and $\dot{T}'$ are of the same sign, then either there is heating where it is warm or cooling where it is cold, and eddy available potential energy (i.e., the variance of temperature within latitude circles) is generated. The reverse—heating where cold or cooling where warm—destroys eddy available potential energy.

To quantify the contribution of radiative heating to the generation of eddy available potential energy we use the model-calculated temperature tendency due to longwave and shortwave radiation ($\dot{T}'_{RAD}$). Figure 2 shows the difference of the generation of eddy available potential energy due to radiation, $G_R(P_E)$, between the control and climatological radiation experiment. By design, the temperature and radiative heating anomalies are uncorrelated in the climatological radiation experiment, so $\overline{T' \times \dot{T}'_{RAD}} \approx 0$ (see Figure S2 in Supporting Information S1). Consequently the change in the generation of eddy available potential energy in Figure 2 resembles the distribution of that quantity in the control. The vertical structure of $G_R(P_E)$ is worth calling attention to. In the midlatitudes $G_R(P_E)$ is positive at mid-levels but negative near the surface and tropopause. We also note that there are two regions of strongly negative $G_R(P_E)$ in the northern and southern polar stratosphere; we will return to this in Section 3.3.

Because the climatological radiation experiment does not directly alter the mean climate, we analyze temperature and radiative heating anomalies on the scale of the cyclone (e.g., Hsieh et al., 2020) rather than in the zonal mean (e.g., Gertler & O'Gorman, 2019; O'Gorman & Schneider, 2008). To visualize these anomalies we generate composites of low-pressure systems in the midlatitudes. Low-pressure systems can be identified in a variety of ways (Walker et al., 2020); in this study we elect for the simplest procedure possible and define a low as the local minimum of sea-level pressure in each 20° × 20° box from 30 to 70 N/S. This provides a large sample of lows (~8–10 per day, in each hemisphere), and is sufficient to illustrate a "typical" low-pressure system. We note that there is approximately the same number of lows between the control and climatological radiation experiment (the difference is <0.5%).

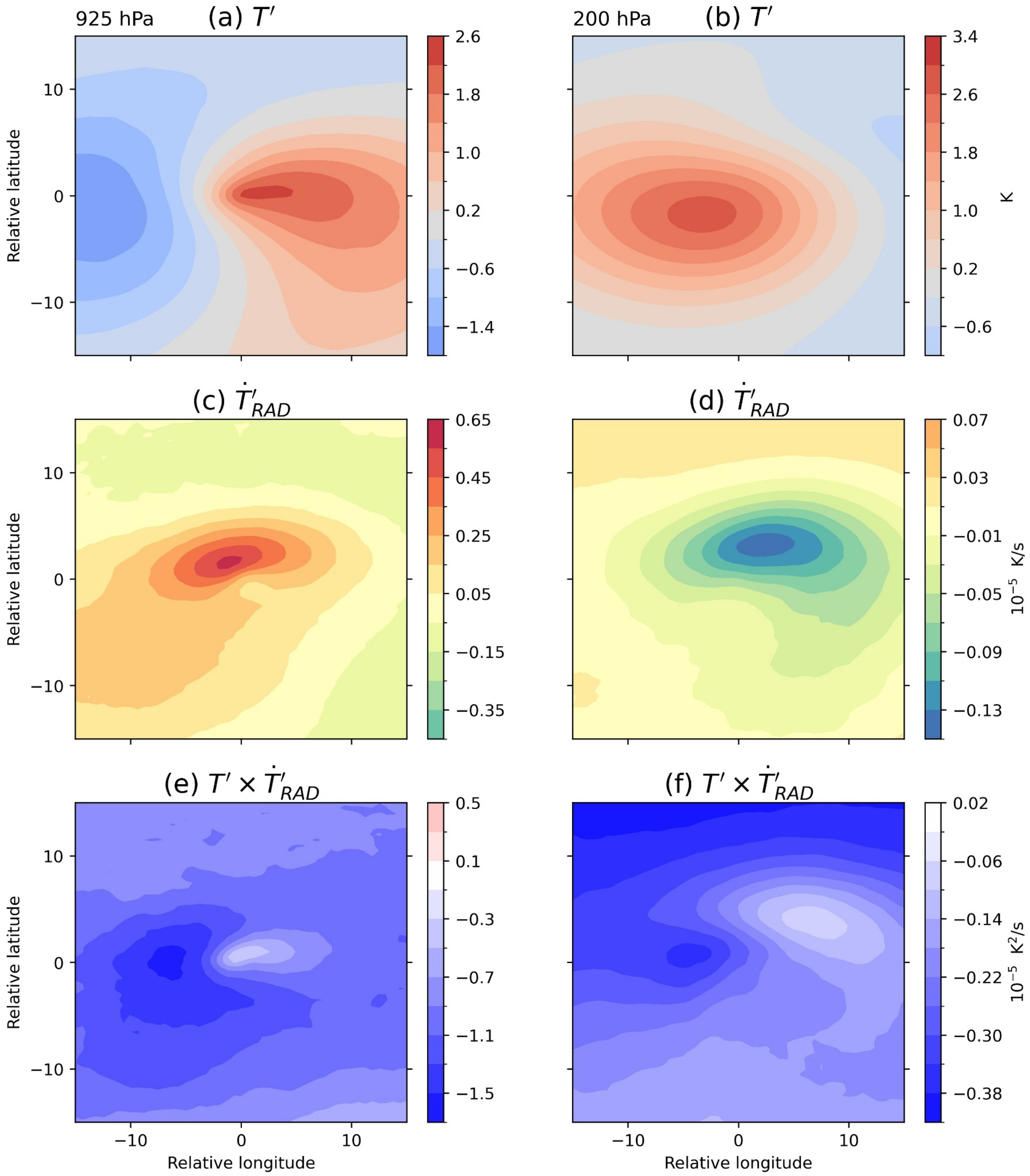

Figure 3. The temperature and radiative heating anomalies in the control, and the covariance between them, in composites of Northern Hemisphere low-pressure systems. The left column is at 925 hPa, the right column at 200 hPa. The covariance between temperature and radiative heating anomalies is negative throughout the domain, indicating that warm air is being cooled and cool air warmed in the control compared to the climatological radiation experiment.

Composites of low-pressure systems in the Northern Hemisphere are reported in Figure 3 on a 30° × 30° domain (the corresponding plots in the Southern Hemisphere are given in Figure S3 in Supporting Information S1). The temperature anomalies (departures from the monthly mean) at (a) 925 hPa and (b) 200 hPa are provided for reference. These show the typical characteristics of midlatitude cyclones, including warm and cold sectors near the surface, separated by fronts, and westward tilt of the temperature anomaly with height. The radiative cooling anomalies are shown in panels c and d. Near the surface there is anomalous cooling in the warm sector and warming in the cold sector, due to the dependence of radiative emission on temperature (the Stefan-Boltzmann law), humidity, and cloud cover. At upper levels there is anomalous cooling, possibly associated with the tops of

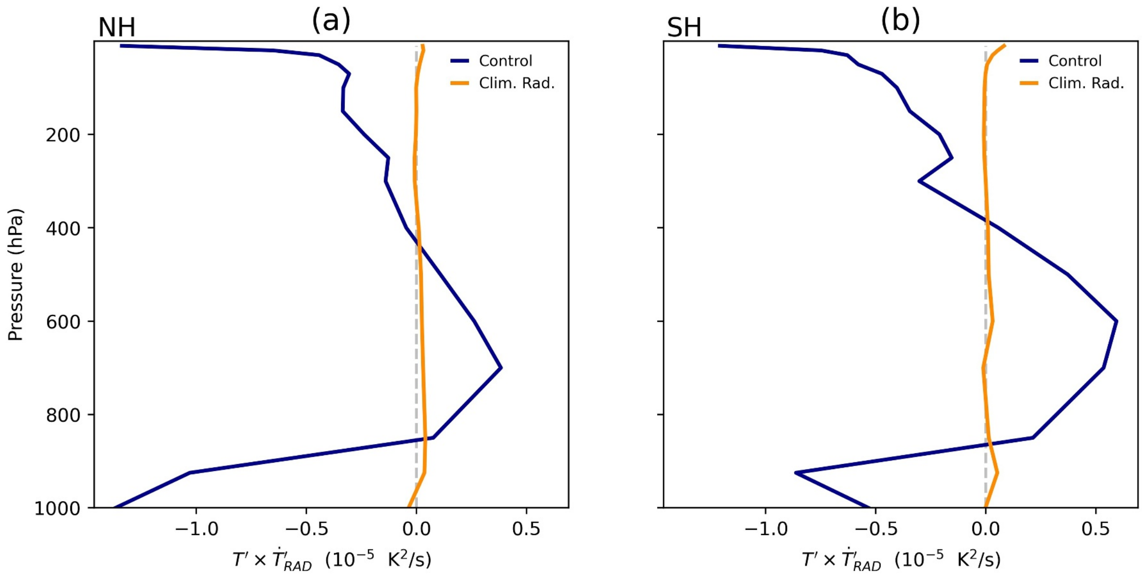

Figure 4. The domain-averaged covariance of temperature and radiative heating anomalies for composites of low-pressure systems. In the control the covariance is positive at mid-levels but negative near the surface and tropopause.

deep clouds forming near the cyclone center. The difference of the covariance of the temperature and radiative heating anomalies, between the control and climatological radiation experiment, is shown in panels e and f. At both 925 hPa and 200 hPa the covariance is everywhere negative, meaning that warm air is cooled and cool air warmed in the control. Finally the domain-averaged $T' \times \dot{T}'_{RAD}$ versus height is shown in Figure 4. In the control the covariance is positive at mid-levels but negative near the surface and tropopause. This supports the idea that radiative heating weakens midlatitude cyclones by destroying eddy available potential energy at upper and lower levels, as suggested by Figure 2.

It is not entirely clear, however, why the destruction of eddy available potential energy near the surface and tropopause seems to prevail over the generation of eddy available potential energy at mid-levels. The change of eddy kinetic energy is concentrated at upper levels in the troposphere, and it is tempting to say that the destruction of eddy available potential energy near the tropopause is responsible for it. However, the relationships between the available potential energy and kinetic energy are only defined for the global average, and not at each location. It is possible that the generation of available potential energy in one part of the atmosphere could alter the kinetic energy in another (Chang et al., 2002). It is also possible that radiative heating could alter other diabatic heat sources, such as latent heat, and thereby indirectly alter midlatitude cyclones (Keshtgar et al., 2023). How the vertical structure of $G_R(P_E)$ affects large-scale eddies remains an open question. We note that a similar idea has been explored with regard to baroclinicity: namely, whether large-scale eddies are more sensitive to changes of the upper- or lower-level horizontal temperature gradient (Catto et al., 2019; Yuval & Kaspi, 2016). Finally, it is worthwhile to compare our results to those of previous studies which have examined the effect of cloud radiative heating on midlatitude cyclones. Using different methods, Schäfer and Voigt (2018), Grise et al. (2019) and Keshtgar et al. (2023) found conflicting responses of the eddy kinetic energy to suppressed cloud radiative interactions. Voigt et al. (2023) attempted to reconcile these results by showing the opposing influences of low and high clouds in cyclone intensification. They argued that the cooling of low cloud tops weakens cyclones, while the cooling of high cloud tops strengthens them, by modifying the static stability near the boundary layer and tropopause. Our results, by contrast, indicate that radiative cooling at upper levels in the warm sector (likely by high clouds) destroys eddy available potential energy, and that radiative cooling at mid-levels in the cold sector (possibly by low clouds) generates eddy available potential energy (see Figures S4 and S5 in Supporting Information S1). Future work in which we explicitly separate the cloud and clear-sky components of radiative heating may help to resolve this apparent contradiction and reconcile the energetic approach of this paper with the momentum approach favored in prior studies.

3.3. The Role of the Stratosphere

In Figure 2 there can be observed in the northern and southern polar stratosphere two regions where the covariance of temperature and radiative heating anomalies is strongly negative. It is conceivable that the loss of eddy available potential energy in one part of the atmosphere could affect the eddy kinetic energy in another, since the two forms of energy are only related when integrated over the entire mass of the atmosphere, and not at each location. Two questions follow: what causes the regions of negative $G_R(P_E)$ in the polar stratosphere, and what effect do they have on the kinetic energy of the troposphere?

The stratosphere is thought to exist in "radiative-dynamical equilibrium": low-wavenumber Rossby waves from the troposphere propagate vertically into the stratosphere, depositing their momentum and inducing a meridional circulation which leads to sinking motion and adiabatic warming (Butchart, 2022). This is the process behind "sudden stratospheric warmings." These warm events cool, in part, by enhanced thermal emission of radiation (the Planck feedback); and therefore positive temperature anomalies in the stratosphere are associated with negative radiative heating anomalies, leading to the negative covariance ($G_R(P_E) < 0$) observed in Figure 2. In the climatological radiation experiment, however, the radiative cooling rate in the stratosphere is fixed, so transient warm events are unable to cool off by the Planck feedback. This leads to more persistent warm events in the climatological radiation experiment, and a higher mean temperature in the polar stratosphere, particularly in winter in the Northern Hemisphere, when vertical propagation of planetary waves is most prevalent (Figure S6 in Supporting Information S1).

Because the troposphere and stratosphere are coupled, it is conceivable that an alteration to the stratosphere in the climatological radiation experiment could affect tropospheric eddies (Kushner & Polvani, 2004). We therefore perform a second experiment, in which the radiative cooling profile is fixed to the monthly mean only in the troposphere (Figure S7 in Supporting Information S1). To demarcate the troposphere from the stratosphere, we define a monthly-varying tropopause as the height above 400 hPa where the lapse rate falls to 2°C/km, based on the definition from the WMO (1957). In the troposphere-only climatological radiation experiment we find an increase of the globally-averaged eddy kinetic energy by ~4% compared to the control. This is slightly smaller than the ~6% increase found in the full-column climatological radiation experiment, and the difference arises mainly in the midlatitudes in the Northern Hemisphere (Table S1 in Supporting Information S1; Figure S8 in Supporting Information S1). This finding suggests a role for the stratosphere in setting the kinetic energy of tropospheric eddies, particularly in the Northern Hemisphere; however, the details of this mechanism are at this point uncertain.

4. Conclusions

Using a high-resolution general circulation model we have shown that, when atmospheric radiative heating is fixed to its climatology, the globally-averaged eddy kinetic energy is enhanced by ~6%. This supports the findings of Schäfer and Voigt (2018) and suggests that, unlike tropical cyclones, midlatitude cyclones are weakened by radiative interactions. It has been the purpose of this study to develop a framework for understanding this result, based on the concept of available potential energy put forward by Lorenz (1955). Our picture of the effect of radiative heating on midlatitude cyclones can be summarized as follows. In a cyclone's warm sector there is convergence of moisture and rising motion leading to deep clouds, while in the cold sector the atmosphere is dry and contains low clouds. In the warm sector there is anomalous radiative cooling, and in the cold sector anomalous radiative heating, near the surface and tropopause. This cooling of warm air and heating of cold air causes a destruction of eddy available potential energy, which offsets the generation of eddy kinetic energy by baroclinic instability. We suggest that the destruction of eddy available potential energy by radiative cooling explains the weakening of midlatitude cyclones when radiative interactions are permitted. Unlike previous studies, which have related the eddy kinetic energy to the mean horizontal temperature gradient or mean available potential energy (O'Gorman & Schneider, 2008), we have shown that radiation weakens the horizontal temperature gradient on the scale of the cyclone.

We have, throughout this article, adopted an "energy viewpoint" for understanding the effect of atmospheric radiative heating on large-scale eddies in the midlatitudes. However, that is only one perspective. Another viewpoint, such as that of momentum (i.e., potential vorticity), is likely to offer complementary insights. How the vertical structure of temperature and radiative heating anomalies affects midlatitude cyclones, and how clouds contribute to radiative heating anomalies, are questions that may benefit from such another perspective.

Data Availability Statement

The HiRAM simulations were performed on Princeton University Research Computing systems. The zonal-mean eddy kinetic energy data set, the zonal-mean generation of eddy available potential energy data set, and the covariance between temperature and radiative heating anomalies data set are available at Mischell (2024).

Acknowledgments
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